CMPE – 211

Preliminary Work (Pre-Lab Activity)

Laboratory Experiment #4

Textbook Material:	Chapter 7	*«Single-Dimensional Arrays»*
	Chapter 8	*«Multidimensional Arrays»*

• • • TASK 1

Write a Java method that returns a new array by eliminating the duplicate values in the array using the following method header:

```
public static int[] eliminateDuplicates(int[] list)
```

Write a Java test Program that reads in ten integers, invokes the method, and displays the result. Here is the sample run of the program:

```
Enter ten numbers: 1 2 3 2 1 6 3 4 5 2 ↵Enter
The distinct numbers are: 1 2 3 6 4 5
```

• • • TASK 2

Write a Java method that merges **two sorted lists** into a new **sorted list**.

```
public static int[] merge(int[] list1, int[] list2)
```

Write a Java test Program that prompts the user to enter two sorted lists and displays the merged list. Here is a sample run. Note that the first number in the input indicates the number of the elements in the list. This number is not part of the list.

```
Enter list1: 5 1 5 16 61 111 ↵Enter
Enter list2: 4 2 4 5 6 ↵Enter
The merged list is 1 2 4 5 5 6 16 61 111
```

• • • TASK 3

Write a Java method that returns the location of the largest element in a two-dimensional array.

```
public static int[] locateLargest(double[][] a)
```

The return value is a one-dimensional array that contains two elements. These two elements indicate the row and column indices of the largest element in the two-dimensional array. Write a Java test Program that prompts the user to enter a two- dimensional array and displays the location of the largest element in the array.

Here is a sample run:

```
Enter the number of rows and columns of the array: 3 4 ↵Enter
Enter the array:
23.5 35 2 10 ↵Enter
4.5 3 45 3.5 ↵Enter
35 44 5.5 9.6 ↵Enter
The location of the largest element is at (1, 2)
```

● ● ● TASK 4

Given a set of cities, the central city is the city that has the shortest total distance to all other cities. Write, Compile and Execute a Java program that prompts the user to enter the number of the cities and the locations of the cities (coordinates), and finds the central city and its total distance to all other cities.

```
Enter the number of cities: 5 ↵Enter
Enter the coordinates of the cities:
  2.5 5 5.1 3 1 9 5.4 54 5.5 2.1 ↵Enter
The central city is at (2.5, 5.0)
The total distance to all other cities is 60.81
```

● ● ● Sources

- Introduction to Java Programming and Data Structures, Comprehensive Version, 11th Edition, Y. D. Liang, Pearson, 2019.